

- 7 (a) Epistasis occurs when a gene at one locus can affect the expression of a gene at another locus.

Define the terms gene and locus.

gene

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locus

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[2]

- (b) Fur colour in mice, *Mus musculus*, is determined by a number of genes. One example is the result of epistatic interaction between two genes, **A** and **B**.

- Allele **A** codes for the production of pigment in the fur.
- Allele **a** does not code for the production of pigment and results in white fur (albino).
- Allele **B** codes for the production of brown fur.
- Allele **b** codes for the production of black fur.

Construct a genetic diagram to show the results, including the ratio, of a cross between two mice heterozygous for both genes.

parent genotypes **AaBb** x **AaBb**

parent phenotypes

gametes

ratio..... [6]

(c) White fur is due to a mutation of the *TYR* gene. This is called albinism.

Explain how a mutation of the *TYR* gene can result in albinism.

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..... [3]

[Total: 11]